EvoOR-Agent: Co-evolving Agent Architectures and Interpretable Reasoning for Automated Optimization

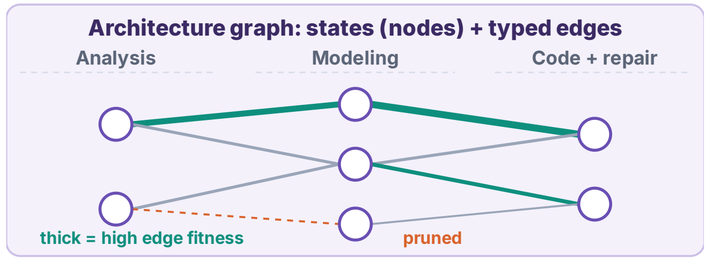
Jiahao Huang, Peilan Xu, Xiaoya Nan, Wenjian Luo · Nanjing Univ. of Information Sci. & Tech. · HIT Shenzhen · arXiv 2604.17708v2

HarnessRSI 81 ★★★

evidence: retained

Paper

1 What changes, what is kept



1 Population of 10 workflows
each an executable OR agent; half of them seeded with the knowledge base

2 Next generation (8×)
elite 20% kept · mutation 50% · recombination 30%: weighted walks on the graph

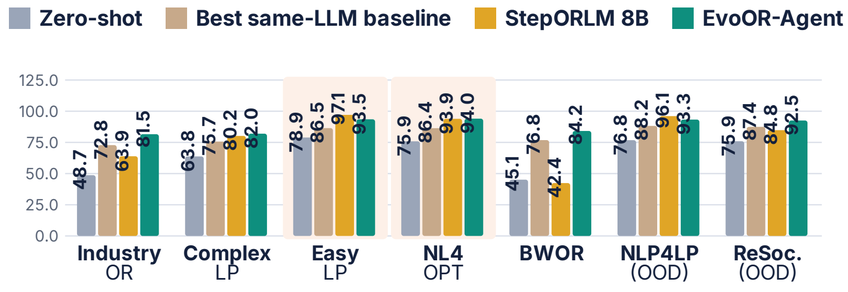
8×

3 Run on 120 training items
solver output within 10^{-3} of the true objective → weighted accuracy (fitness)

4 Update the graph
merge new paths, move edge weights toward fitness (α = 0.5), prune below 0.1

best training fitness → test split
+ 2 benchmarks never used in evolution
LLM weights never change

2 Headline: accuracy %, DeepSeek-v3.2



Best same-LLM baseline: EvoAgent on IndustryOR, OR-LLM-Agent elsewhere. Shaded: EasyLP, NL4OPT (StepORLM higher or level).

3 Controls that decide if the gain is real

Ablation · weighted accuracy (Table V)



AOE step ≈ text-evolution step (paper: "largest component-level gain"; cumulative, order-dependent). +AOE beats every same-LLM baseline, but not StepORLM on EasyLP, NL4OPT, NLP4LP, ComplexLP. Means only.

Budget & protocol

- 400,000-token cap for all evolutionary methods (EvoOR spends ≈410.7K); actual budgets of the baselines not reported
- 120 train / 863 test; **NLP4LP, ReSocratic never evolved on**. Population 10 and generation 8 were read off **test-split curves**; Table III vs test-split values differ slightly
- KB lowers NLP4LP 94.62 → 93.28; its worked example derives from OR-LLM-Agent, and the sanitization prompt names 4 benchmarks, not NLP4LP / ReSocratic
- Population 6 → 10: test 81.2 → 86.8; 10 → 20: 86.8 → 87.4, tokens 410.7K → 812.3K

Tokens / problem (all LLM rows): zero-shot 1,586–1,714 · EvoPrompt 3,394–3,512 · OR-LLM-Agent 3,821–4,096 · EvoAgent 4,673–5,023 · **EvoOR 5,023–5,334**; evolution cost inclusion not stated

4 Where the gain is absent or thin

- **StepORLM (8B fine-tuned) wins:** EasyLP 97.06 vs 94.19 (best EvoOR row); NLP4LP 96.06 vs 93.28 (GPT-5 row 97.75); ComplexLP 80.18 vs 75.53 / 74.89 / 68.82 (GPT-5 / Gemini / Qwen); NL4OPT 93.89 vs 94.03 (DeepSeek, within SD), above the other 3 rows
- Ahead of the best same-LLM baseline in **27 of 28** cells‡; **Gemini ComplexLP 74.89 vs 75.93**; Qwen ComplexLP 68.82 vs 68.03 (SD 1.75 / 1.11)
- Small benchmarks: IndustryOR 42, BWOR 82, ComplexLP 111 items (whole-benchmark sizes)
- After generation 8: test 86.8 → 86.3, train 87.4 → 88.7 (overfitting)

Takeaway

The graph step adds about as much as text-level evolution and holds on two never-evolved benchmarks, under a nominal 400K-token cap. Not uniformly best: a fine-tuned 8B model wins several columns, and test curves set population size and generations.

Our read · HarnessRSI

Score 81 · ★★★ · evidence level: retained

Reports: train/test split, 2 OOD benchmarks, 10 runs, token cap, ablation.

Does not report: graph-update ablation, KB leakage audit, public code; knee / generation 8 chosen on test curves.